

YUGESH S

sivayugesh90@gmail.com | +91-9385996454 | [LinkedIn](#) | [GitHub](#)



PROFESSIONAL SUMMARY

Electrical and Electronics Engineering graduate (8.90 CGPA, Rank 1) with proven hands-on expertise in hardware prototyping, IoT system integration, and power distribution. Skilled in bridging the gap between physical hardware—including microcontrollers, relays, and industrial actuators—and cloud-based telemetry platforms. Focused on applying cross-functional integration skills to industrial automation, SCADA systems, and sustainable energy management solutions.

EDUCATION

V S B College of Engineering Technical Campus | Coimbatore, India *Expected 2027*

Bachelor of Engineering (B.E) in Electrical and Electronics Engineering CGPA: 8.90 (up to 6th semester)

Sri Sankara Vidyalaya Matriculation Higher Secondary School

Higher Secondary Certification Percentage: 87%

CORE COMPETENCIES & SKILLS

- **Core Engineering:** Power Electronics, Control Systems, Circuit Wiring & Assembly, Voltage Regulation (Buck/Boost), Relay Interfacing
- **Automation & Technologies:** IoT System Integration, Microcontroller Interfacing (ESP32, NodeMCU, Arduino), PLC & SCADA Fundamentals, Modbus
- **Programming & Protocols:** C, Java, MATLAB, I2C, SPI, UART, PWM
- **Tools & Instruments:** Arduino IDE, VS Code, Blynk IoT Platform, Multimeters, Oscilloscopes

PROFESSIONAL EXPERIENCE

VVDN Technologies | Pollachi, India *Sept 2024 – Oct 2024*

Intern - Production & Assembly Operations

- Assigned to oversee a commercial hardware assembly line focusing on the physical production and quality assurance of laptops
- Managed and coordinated a team of assembly technicians, ensuring strict adherence to standard operating procedures (SOPs) for hardware mounting, internal wiring, and final casing
- Maintained daily production targets and workflow efficiency while gaining practical exposure to high-volume electronics manufacturing and factory floor operations

TECHNICAL PROJECTS

Smart Stick and Glasses for Visually Impaired | Oct 2025

- Designed a multi-mode mobility aid to provide real-time spatial feedback for visually impaired users across varying environments
- Connected dual ESP32 microcontrollers to synchronize a walking stick and wearable glasses, utilizing the Blynk platform to establish wireless communication and user controls
- Delivered a functional assistive device featuring three distinct sensitivity modes (Indoor, Outdoor, Partial), validating the fusion of cloud telemetry and physical haptic hardware

IoT Smart Irrigation & Automated Pumping System | Dec 2024

- Engineered a remote-controlled irrigation architecture to manage agricultural water distribution based on live soil conditions
- Interfaced a NodeMCU (ESP8266) with a moisture sensor and relay module, configuring a dedicated IoT link for continuous mobile monitoring
- Commissioned a dual-mode setup capable of both autonomous sensor-driven pumping and manual override via a centralized smartphone interface

Fire Fighting Bot Hardware Prototype | Nov 2024

- Constructed an autonomous robotic chassis programmed to track and extinguish localized fire sources using directional thermal inputs
- Executed complex circuit wiring to combine an Arduino UNO control system with an L298N motor driver, a 3-channel IR flame sensor array, a servo, and a relay-actuated DC pump
- Solved multi-voltage power distribution challenges by utilizing a step-down buck converter to safely regulate a dual-cell battery pack down to 3V for the pump, ensuring stable continuous operation.

CERTIFICATIONS & ACHIEVEMENTS

- Patents: Co-Inventor, "Smart Temperature and Gas Detection System" (Indian Patent App. No. 202641034147 A, Published Mar 2026).
- Certifications: NPTEL Elite plus Silver in Internet of Things (IoT), Industrial IoT (Industry 4.0), Cloud Computing, and Soft Skills; Infosys Springboard Basics of C Programming, Python Programming
- Academic Excellence: Department Rank 1 Holder (Semesters 1 to 6) in Electrical and Electronics Engineering
- Leadership: Technical Club Core Leader, mentoring 50+ junior students and leading peer circuit debugging sessions
- Conferences & Hackathons: Smart India Hackathon Participant (presented "Thermolytic" project, Sept 2025); IEEE Conference Presenter (validated high-frequency circuit design for a Wireless Power Transfer paper, Jun 2025)

CURRENT LEARNING & TECHNICAL INTERESTS

- **Industrial Automation:** Currently completing a self-directed study path in PLC architecture and basic Ladder Logic programming to understand edge control in manufacturing environments
- **SCADA & Telemetry:** Researching the integration of IoT telemetry with traditional SCADA systems using industry-standard communication protocols like Modbus.